

Quadratic equations



1 The equation $x^2 - 144 = 0$ has a positive integer solution of $x = 12$. Find its other solution.

2 Solve the following equations:

a $x^2 = 2$

b $x^2 = 25$

c $x^2 = 121$

d $x^2 = 294$

e $x^2 - 121 = 0$

f $x^2 - 10 = 15$

g $\frac{x^2}{16} - 2 = 2$

h $\frac{x^2}{25} - 3 = 6$

i $(x + 3)^2 = 49$

j $(x - 3)^2 = 64$

k $(x - 6)^2 = 2$

l $(2 - x)^2 = 81$

m $(x - 7)^2 = 81$

n $(7 - x)^2 = 81$

o $(8x + 9)^2 = 256$

p $81x^2 - 16 = 0$

3 Solve the following equations:

a $x(x + 7) = 0$

b $x(2x - 9) = 0$

c $(10x - 9)^2 = 0$

d $(4x - 9)^2 = 0$

e $(-3 + 7x)^2 = 0$

f $(x - 4)(x - 2) = 0$

g $(x - 6)(x + 7) = 0$

h $(8x - 5)(3x - 7) = 0$

i $(3x + 8)(5x - 7) = 0$

j $(3x - 17)(2x - d) = 0$

4 Solve the following equations:

a $4y^2 = 100$

b $25y^2 = 36$

c $-3k^2 = -12$

d $81k^2 + 8 = 24$

e $-25v^2 + 64 = 0$

f $10(p^2 - 7) = 930$

g $4m(m + 5) = 0$

h $\frac{m}{2}(m + 5) = 0$

5 Solve the following equation for x , in terms of a and c . Assume a and c are positive.

$$ax^2 - c = 0$$

Applications

6 The equation $4x^2 + kx + 16 = 0$ has one solution: $x = 2$. Find the value of the coefficient k .

7 The equation $ax^2 - 32x - 80 = 0$ has one solution: $x = 4$. Find the value of the coefficient of a .

- 8 The Widget and Trinket Emporium has released the forecast of its revenue over the next year. The revenue R (in dollars) at any point in time t (in months) is described by the equation:

$$R = -(t - 12)^2 + 4$$

When will the revenue be zero?

- 9 Neville needs a sheet of paper x cm by 13 cm for an origami giraffe. The local origami supply store only sells square sheets of paper.

The lower portion of the image below shows the excess area A of paper that will remain after Neville cuts out the x cm by 13 cm piece. The excess area, in square centimeter, is given by the equation:

$$A = x(x - 13)$$

- a At what lengths x will the excess area be zero?
- b For what value of x will Neville be able to make an origami giraffe with the least amount of excess paper?

